A 161-mW 56-Gb/s ADC-Based Discrete Multitone Wireline Receiver Data-Path in 14-nm FinFET

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***Abstract*— This article introduces a wireline receiver (RX) data-path employing discrete multi-tone (DMT) modulation for communicating over electrical links. The DMT RX incorporates a fully digital equalization data-path, with a synthesized and automatically placed and routed digital signal processor (DSP) following a 10-bit time-interleaved pipelined successive-approximation register analog-to-digital converter (TI-PISAR ADC). The prototype RX chip implemented in a 14-nm FinFET process demonstrates a lane data rate of 56 Gb/s dissipating 161 mW including the ADC and the DSP power. The energy efficiency of 1.2 pJ/b for the DSP and 2.9 pJ/b for the entire RX was achieved with the data-rate of 56 Gb/s for communicating over channels exhibiting up to 28-dB loss at 14 GHz with a bit-error-rate (BER) better than 2e-4.**

***Index Terms*— Discrete multi-tone (DMT), inter-symbol interference (ISI), orthogonal frequency-division multiplexing (OFDM), receiver, SerDes, serial-data transceiver, wireline.**

I. Introduction

THE ever-increasing demands on higher interconnect bandwidth in datacenters and high-performance computing systems have been pushing the per-lane data rate to 56 Gb/s and further. In order to double the current 28-Gb/s data rate without replacing pre-installed copper cables for minimizing the infrastructure upgrade cost, the bandwidth efficiency must be increased in order to avoid a dramatic increase in signal attenuation at Nyquist frequency. This motivated the >32 Gb/s electrical I/O transceiver (TRX) design trend from non-return-to-zero (NRZ) to four-level pulse-amplitude modulation (PAM-4) [1], [2].

Furthermore, taking advantage of technology scaling, implementation of equalizers as a digital signal processor (DSP) following analog-to-digital converter (ADC) is becoming more and more popular [3]–[10]. ADC-based PAM-4 receiver (RX) prototypes showed their strong equalization capabilities compensating for the loss of up to around 40 dB at 14 GHz [10], [11]. Modern CMOS process technology allows the RX DSP to be equipped with a large number of feed-forward equalizer (FFE) taps (14–19-tap [6], [7], [10]) for channel-inversion, canceling the post-cursors together with one-tap or two-tap speculative decision-feedback equalizer (DFE). Still, the digital PAM-4 RXs are typically more power-hungry as compared to analog RXs due to the complex DSP that is needed to compensate for the high attenuation at f_{Nyq}. Furthermore, while the mid-resolution ADC is no-longer a power bottleneck of a digital RX in advanced process nodes, its power consumption at tens of gigasamples per second still occupies an excessive portion in overall RX power budget.

By increasing the bandwidth efficiency of the signal given a target data rate, the power consumed in the digital-to-analog converter (DAC) and the ADC can be reduced, thanks to the reduced required sampling rate. A straightforward approach to increase the number of bits-per-symbol is to increase the size of the constellation, e.g., from PAM-4 to PAM-8; however, this is not adequate in time-domain >56 Gb/s RX due to the excessive DSP equalizer complexity, especially that of the speculative DFEs, which increases exponentially in the constellation order. Analog multi-tone signaling can potentially provide the bandwidth efficiency better than that of the baseband PAM-4, especially in multi-drop bus interfaces [12]–[15], but requires multiple clocks for orthogonal local oscillator frequencies, which limits the total number of frequency-domain sub-channels.

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This article describes a bandwidth- and energy-efficient RX implementation employing discrete multi-tone (DMT) modulation for serial data communication at 56-Gb/s data-rate over channels with up to 28-dB attenuation at 14 GHz [16]. Although prior arts such as in [17] and [18] showed the feasibility of DMT modulation for 56–112-Gb/s optical links, this work first demonstrates a greater than 56-Gb/s DMT-based electrical wireline RX silicon that integrates the ADC and the DSP with its energy efficiency better than 3 pJ/b. The 56-Gb/s all-64-quadrature amplitude modulated (QAM)-DMT signal occupies a bandwidth of 11.2 GHz which requires only a 22.4-GS/s DAC and ADC sampling rate, while 28 GS/s is required in a PAM-4 TX/RX for the same data rate. The data-path is designed as a fully digital circuit to better exploit the digital-friendly nature of the FinFET process. Several techniques are applied for an area- and power-efficient DSP data-path implementation, which is detailed in Section III.

The rest of this article is organized as follows. Section II briefly reviews DMT modulation with a special emphasis on its application to electrical serial links. Section III describes the system overview and the detailed implementation of the DMT RX. Measurement results of the fabricated prototype are presented in Sections IV. Section V concludes this article.

II. Discrete Multitone Modulation in Wireline Serial Link

Fig. 1(a) illustrates a simplified DMT TX, in which N is an even number representing the number of frequency bins (tones) of the inverse discrete Fourier transform (IDFT) processor and X_k with $k \in \{0, 1, \ldots, N/2 - 1\}$ represents frequency-domain QAM symbols. For transmitting data, complex-numbered QAM symbols are first created in frequency-domain from a raw binary data stream. As illustrated in Fig. 1(b), a binary stream is split into groups of bits, which are then mapped to corresponding QAM constellation points defined as I/Q-complex values. For example, 10 bits can be split into a group of 6 bits and a group of 4 bits. Then, a 64-QAM symbol can be created by mapping 3 bits to the I-phase and 3 bits to the Q-phase, and a 16-QAM symbol can be created by mapping 2 bits to the I-phase and 2 bits to the Q-phase. The size of the groups can be different, resulting in different QAM orders for each IDFT frequency bin, as shown in Fig. 1(a). It is worth noting that the IDFT output should be purely real-valued as the TX output signal should be transmitted toward the transmission line in the form of an electrical signal without analog upconversion and I/Q mixing. Hence, the first half and the second half of the IDFT input values should satisfy the Hermitian symmetry (complex conjugate symmetric inputs) to guarantee a purely real-valued IDFT output. This means that only one-half of the IDFT frequency-domain input tones actually carry information, while the other half is created by simply copying and complex-conjugating the first half.

The IDFT processor then transforms the frequency-domain symbols into time-domain samples, where the N output samples of an IDFT are defined as a DMT symbol. To protect a DMT symbol from inter-symbol interference (ISI) caused by

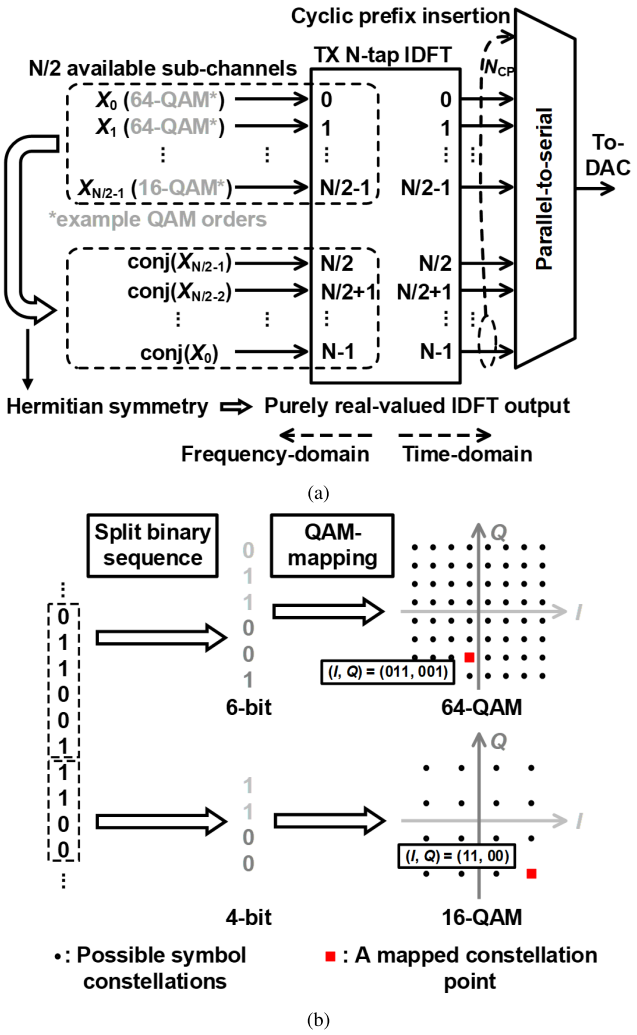


Fig. 1. Simplified illustration of (a) DMT TX and (b) binary-to-QAM mapping in a DMT TX.

the channel delay spread, a cyclic prefix (CP) is inserted by prepending the last N_{CP} samples in the DMT symbol before its beginning [19], at the cost of bandwidth overhead of the signal to be transmitted. This way of symbol protection from ISI preserves the sub-carriers' mutual orthogonality, while simple guard-interval insertion does not preserve the orthogonality of the received sub-carriers. The length of CP should be longer than the channel's delay spread in order to completely remove ISI; otherwise, residual ISI can affect the received DMT symbol. The CP length and the amount of the residual ISI depend on the system's raw bit-error-rate (BER) target given other parameters and can be determined by simulation when channel characteristics are defined. The CP followed by the DMT symbol is then transmitted toward the channel by a DAC.

Since the IDFT transforms a frequency-domain signal to a time-domain signal, the IDFT input data symbols are associated with the normalized frequency bins, as shown in Fig. 2. The first IDFT input corresponds to the dc band, and the $(N/2-1)$th bin corresponds to the highest frequency band. Each sample of the time-domain IDFT output signal

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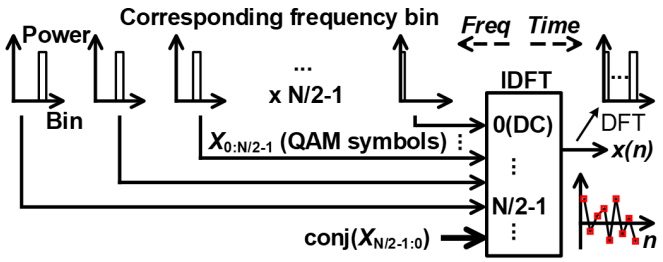


Fig. 2. IDFT input bins and corresponding normalized frequency bins.

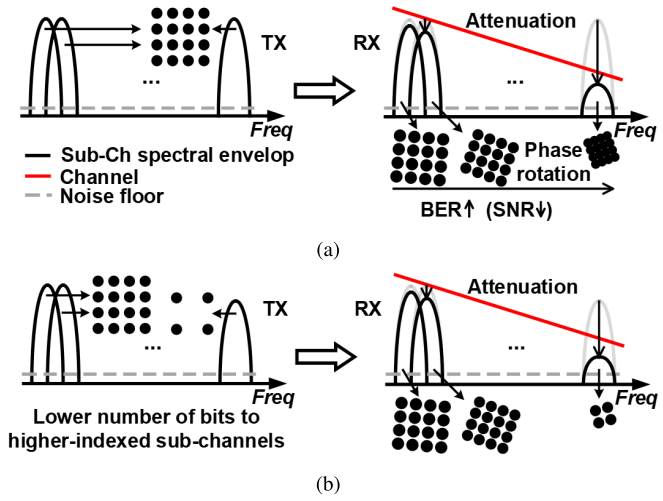


Fig. 3. Different bit-loading on the TX (left column) and the corresponding constellation diagrams after the channel on the RX (right column). (a) Uniform bit-loading and (b) BER-aware bit-loading without taking the quantization noise into account.

includes the weighted sum of all frequency components. The IDFT output (DMT signal) spectrum can be adjusted by configuring the order and/or power of QAM symbols on particular IDFT frequency bins. A uniform bit-loading with a high-order QAM constellation can maximize the bandwidth efficiency. However, in frequency bands with low SNR due to strong channel attenuation [Fig. 3(a)], the error-vector magnitude of the corresponding sub-channel can be significant. Since the BER of these low-SNR sub-channels dominates the BER of the overall system, bit-loading is widely applied in communication systems with DMT/orthogonal frequency-division multiplexing (OFDM), assigning high-order QAM symbols to high-SNR sub-channels and lower-order QAM symbols to low-SNR sub-channels, as illustrated in Fig. 3(b). Due to the larger minimum distance between constellation points of the lower order QAM symbols compared to high-order QAM symbols, a balanced low BER can be maintained on the RX-side. However, in this case, the data-rate depends on the specific channel condition, and finding the optimum bit-loading can be more complicated taking the DAC/ADC conversion rate and their corresponding power consumption, bandwidth, and signal-to-noise-and-distortion ratio (SNDR) into account.

In this article, uniform 64-QAM symbols are loaded onto every sub-channel for reducing the required signal bandwidth for a given target data-rate of 56 Gb/s, with a gradually increasing power loading to partially balance the SNR over the low-to-high-frequency sub-channels. Although DFT with

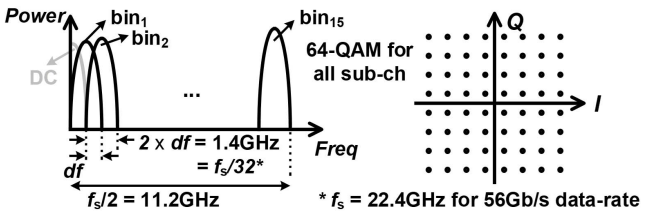


Fig. 4. Frequency planning of the 56-Gb/s DMT signal with uniform 64-QAM sub-channels to minimize the DAC/ADC conversion rate.

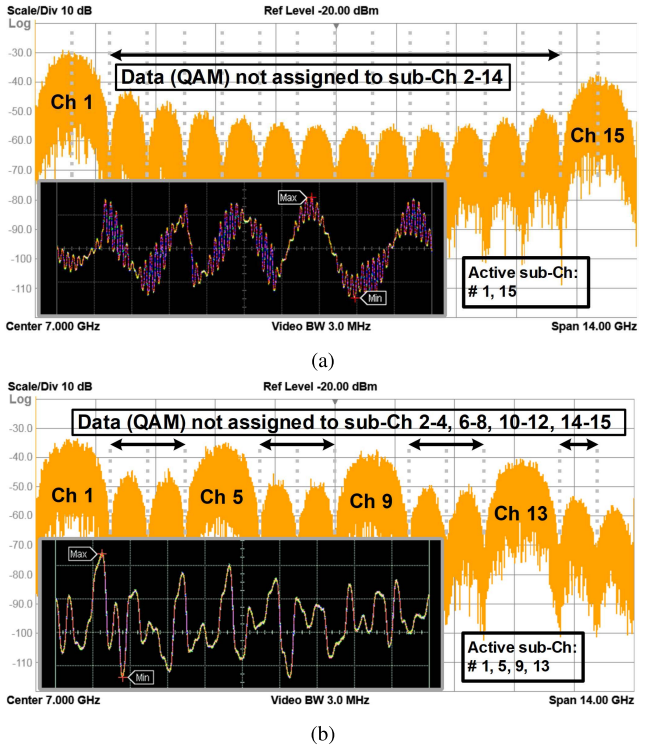


Fig. 5. TX output spectrum of example DMT signals and their time-domain waveform in their inset with (a) 1st and 15th sub-channels enabled and (b) 1st, 5th, 9th, and 13th sub-channels enabled, both with a total of 15 available non-dc sub-channels.

a large number of taps can reduce the bandwidth and sampling rate overhead coming from CP insertion, a 32-tap DFT ($N = 32$) is chosen due to the power and DSP footprint constraints: a 64-tap DFT did not fit in the design space and a hybrid 32-/64-tap DFT was not better than a 32-tap DFT in terms of both power and area. Hence, the number of sub-channels is 15 ($N/2 - 1$) excluding the dc band, and the sub-channel data-rate for the target 56 Gb/s is 3.733 Gb/s. The number of CP is chosen to be 4 based on BER simulation for the worst target channel exhibiting 28-dB loss at 14 GHz. The single-sided bandwidth of a sub-channel spectrum, which is equivalent to the spacing between adjacent 64-QAM sub-channels in frequency-domain taking the CP overhead into account, is given by

$$\frac{56\ \text{Gb/s}}{15} \times \frac{1}{6\ \text{b/Hz}} \times \frac{32+4}{32} = 700\ \text{MHz}. \quad (1)$$

As one sub-channel (dc band) out of 16 available sub-channels of a 32-tap DFT was not used, the overall bandwidth density

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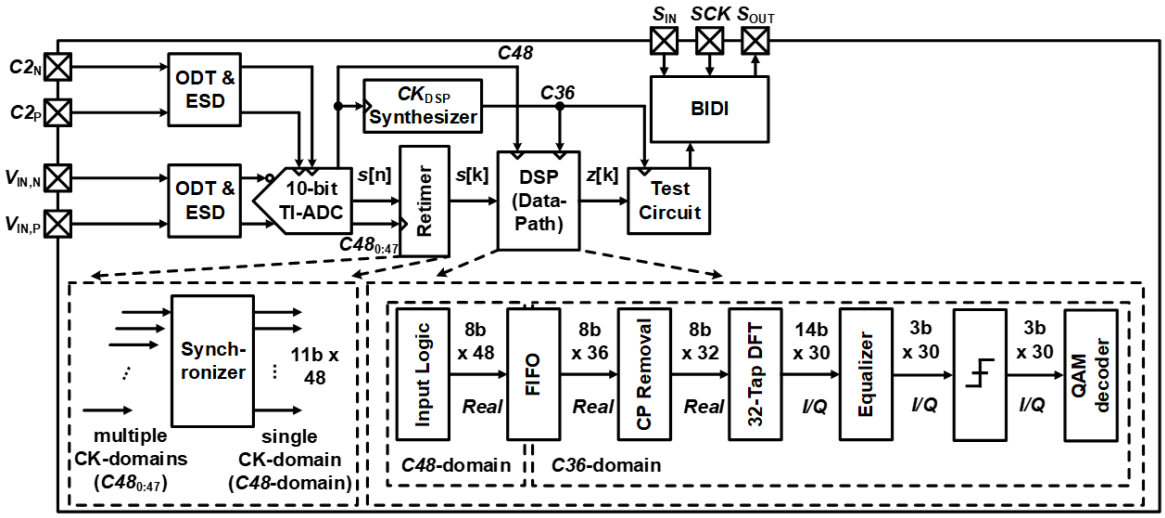


Fig. 6. Block diagram of the implemented RX.

of all-64-QAM DMT symbol without CP is 6 b/Hz × 15 ÷ 16 = 5.625 b/Hz. Hence, taking the CP overhead into account, the conversion rate of the Nyquist data converters should be

$$\frac{56\ \text{Gb/s}}{5.625\ \text{b}} \times \frac{36}{32} \times 2\,\text{S} = 22.4\ \text{GS/s}. \tag{2}$$

The overall DMT signal configuration and the frequency planning are illustrated in Fig. 4.

Fig. 5 shows examples with two different sub-channel configurations. Fig. 5(a) shows a DMT signal configuration with the lowest frequency sub-channel and the highest frequency sub-channel carrying data while the other sub-channels are turned off, and Fig. 5(b) shows a DMT signal configuration with four sub-channels carrying data while 12 other sub-channels are disabled. From the waveform shown in the inset of Fig. 5(a), the high-frequency transition superimposed on top of the low-frequency wave is clearly visible. Yet, for this example, information about the transmitted data cannot be extracted directly by observing the time-domain waveform shown in the inset of Fig. 5(b) as different frequency components are overlaid in the time-domain.

III. Receiver Design

A. Overall System

Fig. 6 shows the block diagram of the implemented RX. The analog block consists of the input network and a 48-way time-interleaved pipelined successive approximation register (TI-PISAR) ADC. The remaining RX is digital. Differential half-rate clock (C2) and data inputs are on-die terminated with 100-Ω differential resistors and T-coils for input bandwidth enhancement. The DSP including the data-path and the measurement circuitry is fully synthesized using standard cell libraries followed by placement and routing with an automatic back-end design flow. The RX does not include a continuous-time linear equalizer (CTLE) before the ADC-stage; hence, the entire equalization data-path is in the DSP. The retimer synchronizes the 48-way time-interleaved (i.e., skewed) ADC output to a single-clock edge so that the DSP receives aligned parallel ADC output of 48 samples at a rate of 1/48 (C48) of the sampling clock. The CK_{DSP} synthesizer synthesizes the clock for the main DSP data-path (explained in the next subsection) and the on-chip measurement circuits, by using eight phases of C48 clock spaced by 45°, with multiply-by-four and divide-by-three circuits. The test circuit can be configured as an on-chip constellation scanner, a BER counter for performance measurement purposes, and as an equivalent-time digital oscilloscope for chip debugging purposes. The BIDI is a bidirectional serial interface for chip programming and measured data collection.

B. DSP Data-Path Implementation

The main DSP data-path consists of seven major functional blocks: an Input Logic, a FIFO, a CP remover, a 32-tap DFT processor, a frequency-domain equalizer (FDE), a sampler, and a QAM decoder, where the first five blocks are detailed in this article since the sampler and the QAM decoder implemented in this article are rather trivial. The Input Logic shown in Fig. 7 is the interface to the ADC. Since the ADC employed in the implemented DMT RX provides 11 raw output bits where 1 bit is redundant [20], the 11-bit code including the redundant bit is converted to a 10-bit binary-weighted code. For a compact data-path, the first eight MSBs are then sent to the following data-path disregarding the last two LSBs. A digital offset calibration is applied individually to all sub-ADC outputs, where the offset values are estimated off-chip. Next, the RX must synchronize to the symbol boundary (i.e., to the first sample of the CP). To this end, the CP index is estimated off-chip, by cross-correlating a known TX sequence with RX samples captured by the on-chip equivalent-time oscilloscope. As this step is required only once after the chip power-on, the CP index estimation hardware would not bring a significant power overhead and can operate in serial, based on a controller, a multiply-accumulate circuit and a memory of 3 × 72 Byte for storing the training sequence, the received sequence, and the cross correlation values, if implemented on-chip. After the estimation of the CP index, multiplexers in the Input Logic are

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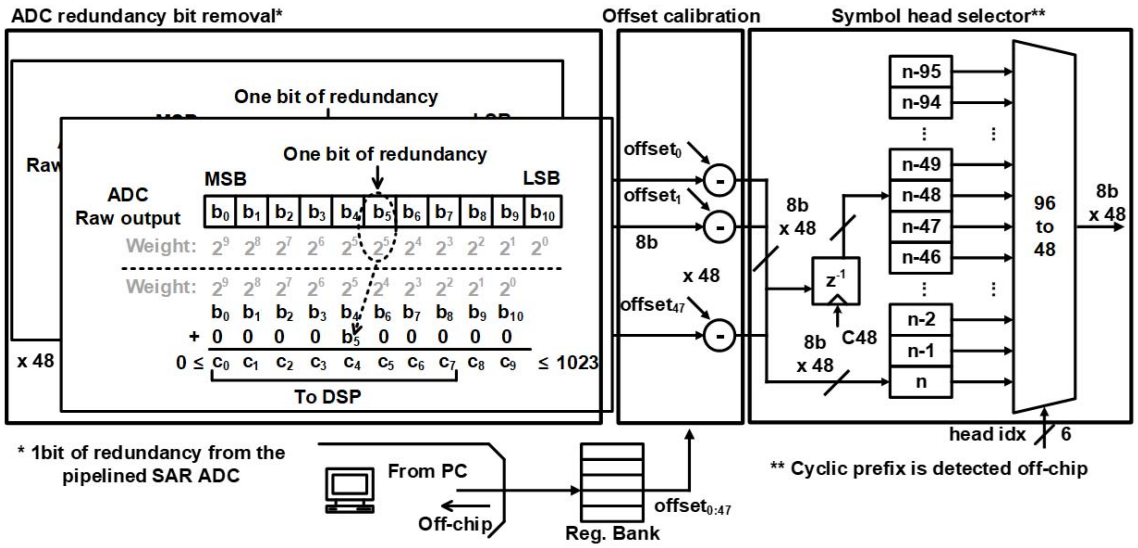


Fig. 7. Input logic layer of the DSP data-path.

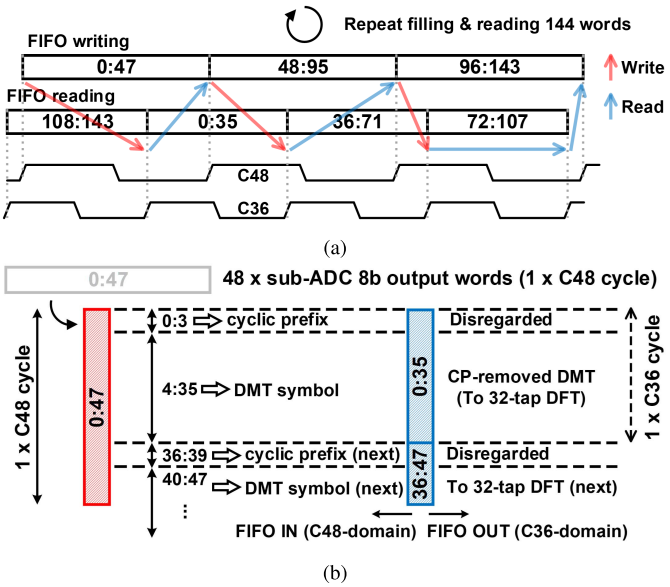


Fig. 8. (a) Timing diagram of the FIFO. (b) Operation of the CP remover.

programmed to remove any integer-sample time offset between TX and RX before passing on the samples to the data path.[1]

As the TI-ADC output is synchronized to a C48 clock flushing 48 words to the DSP each C48 cycle, while the DSP data-path receives 32 samples for the DMT symbol plus $N_{CP} = 4$ CP taps at a rate of 1/36 (C36) of the sampling rate, a FIFO is required to interface the two clock domains. Fig. 8(a) shows the timing diagram of the FIFO. The FIFO has 144 8-bit words, which is the least common multiple of 48 and 36. Packets of 48 samples are sequentially written to the FIFO, and packets of 36 samples are sequentially read from the FIFO.

After the initial time-alignment in the Input Logic, the CP inserted on the TX-side can easily be removed on the RX-side. In Fig. 8(b), the red-colored bar represents a packet of 48 samples as it is written to the FIFO, and the blue-colored bar represents a packet of 36 samples and part of the next packet that is read from the FIFO. The CP removal is done simply by disregarding the first four samples out of the 36 samples read from the FIFO, then the following 32 samples correspond to the DMT symbol. For the next read cycle, in the same way, the first four samples are disregarded for the CP removal, and the following 32 samples correspond to the next DMT symbol.

The recovered time-domain DMT symbol is transformed back to frequency-domain by a 32-tap parallel DFT processor. Given that the DFT input is purely real-valued, it is guaranteed that the output of the DFT is complex conjugate symmetric. Hence, half of the DFT output taps $k \in \{16, 17, \ldots, 31\}$ can be disregarded, as they do not carry additional data information on top of what is carried by the other output tones $k \in \{0, 1, \ldots, 15\}$. A 32-tap parallel DFT with $k \in \{0, 1, \ldots, 15\}$ can be expressed as

$$X_{32}(k) = \sum_{n=0}^{31} x_n \cdot e^{-j\frac{2\pi kn}{32}}$$

$$= \sum_{n'=0}^{15} \left(x_{2n'} e^{-j\frac{4\pi kn'}{32}} + x_{2n'+1} e^{-j\frac{4\pi kn'+2\pi k}{32}} \right) \quad (3a)$$

$$= X_{16,0}(k) + X_{16,1}(k) e^{-j\frac{\pi k}{16}} \quad (3b)$$

where x_n represents a time-domain sample corresponding to the DFT input tap n, $X_{16,0/16,1/32}$ represent frequency-domain samples of two 16-tap sub-DFTs and the 32-tap DFT, respectively, k is the DFT output tap index ranging from 0 to 15, n' is the DFT input sample pair. The first term of (3a) represents the 16-tap DFT with even-numbered inputs, and the second term of (3a) represents the 16-tap DFT is rotated by $(2\pi k/32)$ *rad* with odd-numbered inputs. Re-writing (3a) as (3b), the resulting 32-tap DFT processor considering only the first half of the outputs ($k \in \{0, 1, \ldots, 15\}$) is implemented by two 16-tap

[1]Note that any remaining non-integer sample time offset is later handled inherently by the DMT equalization.

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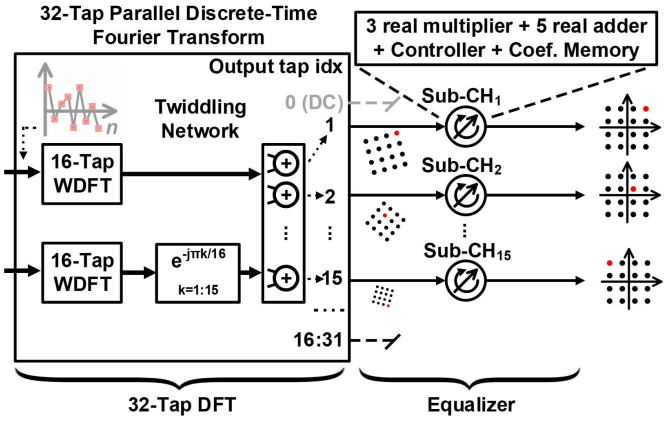


Fig. 9. 32-tap parallel DFT and a column of FDE composed of 15 complex multipliers and control circuitries.

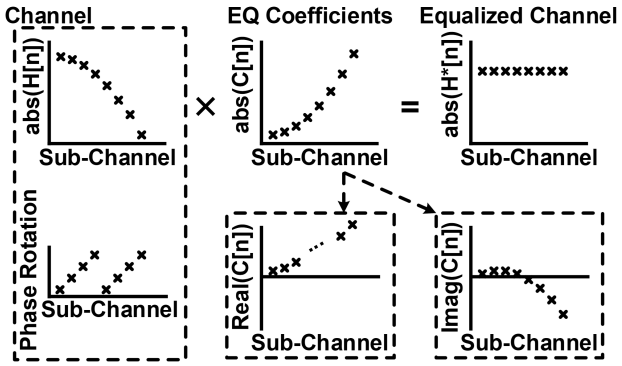
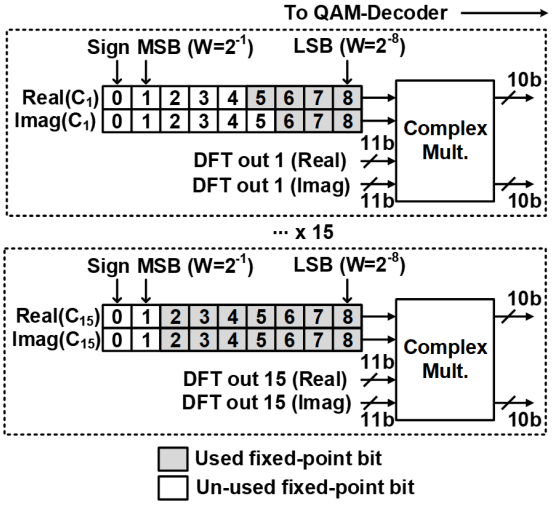
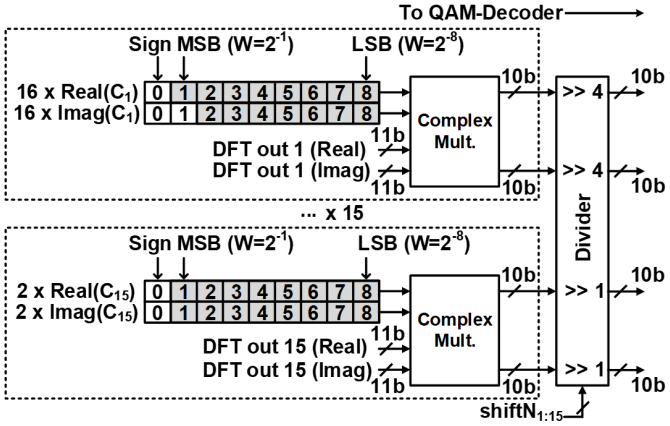


Fig. 10. Frequency-domain equalization process.



(a)



(b)

Fig. 11. FDE's complex multiplier (a) without and (b) with the per-sub-channel coefficient re-scaling.

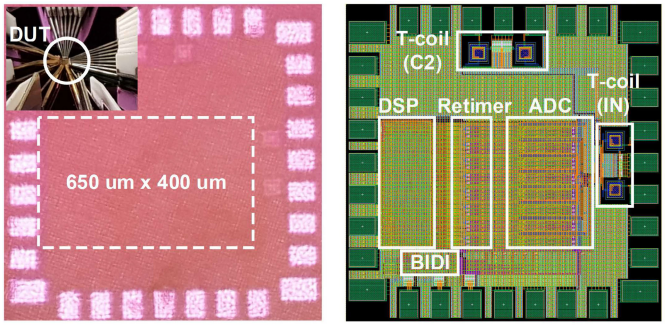


Fig. 12. Die photograph and the layout of the implemented DMT RX.

Winograd DFT processors [21] for a compact area followed by a twiddling network for the 16-tap DFT that receives odd-numbered inputs. Fig. 9 shows the designed 32-tap DFT processor followed by an FDE for per-tone equalization. The constellation diagram at each DFT output tap corresponding to each individual sub-channel is shrunk and rotated by a factor that is different for each tone, due to the frequency-dependent attenuation and the uncompensated non-integer sample delay introduced by the transmission line channel. These phase rotations and attenuation in the frequency-domain are compensated by a single complex-valued multiplier for each sub-channel, with an off-chip coefficient estimation. Since no data are assigned to the dc-tone, a total of 15 complex-valued multipliers are required. The frequency-domain equalization based on element-wise 9-bit complex multiplications is depicted in Fig. 10.

With an exponentially increasing channel attenuation as a function of frequency, elementwise multiplication with exponentially increasing coefficients should be applied to the DMT tones to flatten the channel. Unfortunately, the corresponding factor between the smallest coefficient and the largest equalization coefficients can be significant with lossy transmission lines. As a consequence of this large dynamic-range requirement, the accuracy of a straightforward fixed-point implementation of the equalizer degrades significantly for severely attenuated tones with a strong and detrimental impact on the precision of the phase-rotation, as shown in Fig. 11(a). To address this issue, an individual pre-scaling of the coefficients with a power-of-two factor is applied to each tone. In the example in Fig. 11(b), the equalizer coefficient of the first tone is multiplied by a factor of 2^4, while the coefficient of the last tone is only multiplied by a factor of 2^1 to fully utilize the accuracy of the respective multipliers. The equalized samples at the output of the complex multipliers are then scaled back as appropriate by the same power-of-two with minimal hardware overhead.

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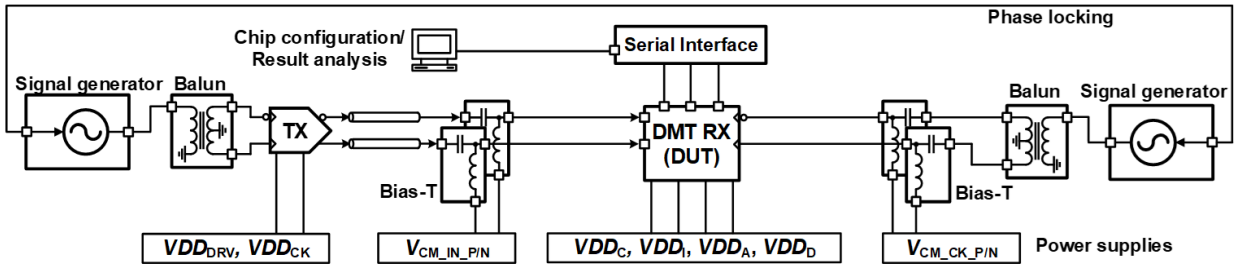


Fig. 13. Measurement setup.

IV. Measurement Results

Fig. 12 shows a die photograph and the layout of the DMT RX prototype implemented in a 14-nm FinFET process. The active area is 650 um × 400 um with a DSP area of 180 um × 400 um. The measurement setup for the DMT RX characterization is shown in Fig. 13. Two external clock sources (Keysight E8257D) are synchronized by a 1-GHz reference clock to generate two 11.2-GHz C2 clocks for the TX and for the RX. The single-ended 11.2-GHz clocks are converted to differential clocks using baluns (Hyperlabs HL9404), both for the DAC and for the ADC. An 8-bit DAC [22] is used as the DMT TX that repeatedly sends a data stream stored in its on-chip memory, where the time-domain DMT stream is created on the PC. The channel includes printed circuit board (PCB) traces, connectors, striplines of the test board (Wildriver ISI-28-32), and cables. The channel output differential signal is ac-coupled and enters to the RX with proper RX ADC common-mode voltage provided by external dc supplies (Agilent E3644). The RX chip is measured on an RF needle probing station, communicating with a PC via a BIDI.

The end-to-end losses of the three different tested channels $CH_{\rm A}$, $CH_{\rm B}$, and $CH_{\rm C}$ at 14 GHz are 11, 18, and 28 dB, respectively. The highest frequency components of the tested 56-Gb/s DMT signal (including a 12.5% of bandwidth overhead due to CP) is 11.2 GHz, where the corresponding attenuation levels are 8, 13, and 20 dB for $CH_{\rm A}$, $CH_{\rm B}$, and $CH_{\rm C}$, respectively. The DMT test patterns are created based on three independent PRBS15 sequences and are eight-tap finite-impulse response (FIR)-filtered in software, for partial channel shortening by de-emphasizing the low-frequency components up to 10 dB. When the TX near-dc components are de-emphasized in digital, the dynamic range of the low-frequency signal amplitude in the digital-domain reduces. Consequently, the low-frequency digital signal uses only a portion of DAC dynamic range, resulting in potentially insufficient conversion SNR for 64-QAM symbols. In contrast, the high-frequency components are boosted compared to the low-frequency components; hence, they use more DAC bits. If the low- and high-frequency power levels are equivalent on the RX-side due to TX de-emphasis together with the channel attenuation, the level of distortion introduced by ADC quantization is equivalent for both low- and high-frequency tones. As shown in Fig. 14(a), with pre-emphasis the relative conversion loss penalty of the first (near-dc) tone is more pronounced than that of the 15th (highest frequency) tone, resulting in a more noisy

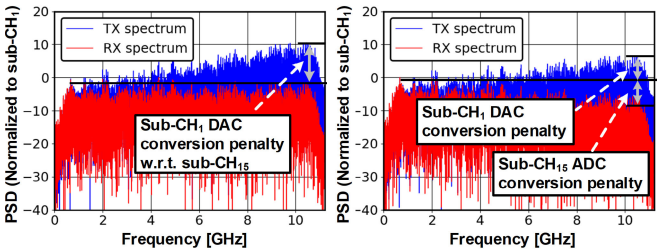


Fig. 14. TX/RX spectrum (a) without and (b) with TX sub-channel gain balancing for balanced quantization SNR at near-dc and high-frequency sub-channels.

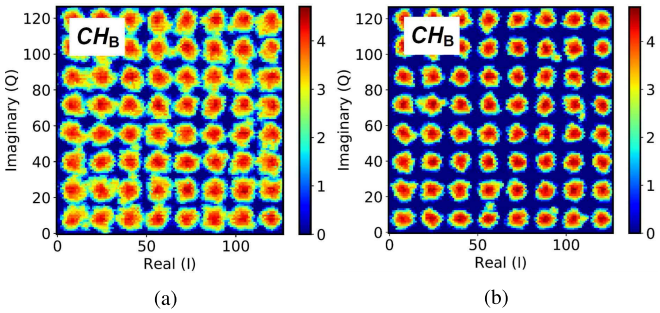


Fig. 15. Equalized constellation diagrams for $CH_{\rm B}$ (a) without and (b) with sub-channels' power adjustment.

recovered constellation diagram for the first tone [Fig. 15(a)]. To alleviate this effect, the power of each tone is adjusted in the digital-domain to equalize the loss across all tones, as shown in Fig. 14(b), which enhances the overall constellation diagram quality as shown in Fig. 15(b).

The received and equalized 56-Gb/s all-64-QAM DMT constellation diagrams for channel $CH_{\rm A}$ and $CH_{\rm C}$ are shown in Fig. 16, together with bathtub curves. The equalized constellation points are more dispersed as the channel becomes more attenuative (from $CH_{\rm A}$ to $CH_{\rm C}$). The bathtub curves for the symbol-error-rate are drawn by overlapping sets of four (2 × 2) constellation points on top of each other for each channel. By scanning the horizontal and vertical cross sections of each overlapped constellation diagram, the symbol-error-rate at optimal decision point for each channel can be measured. A symbol-error-rate of less than 2e-4 was measured for $CH_{\rm C}$, and better than 1e-6 symbol-error-rate was measured for $CH_{\rm A}$ and $CH_{\rm B}$, as shown in Fig. 16(c). Under nominal supplies of 0.75, 0.8, 0.7, and 0.7 V for DSP, clock buffer, TI-ADC interleaved, and ADC, respectively, the total RX power consumption is 161 mW (with 68 mW

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TABLE I

PERFORMANCE SUMMARY OF DMT RX AND COMPARISON WITH RELEVANT PRIOR ARTS

	JSSC17 [4]	ISSCC18 [6]	ISSCC18 [7]	ISSCC19 [10]	**This work**
Data Rate (Gb/s)	56	56	64.375	56	**56**
Modulation	PAM-4	PAM-4	PAM-4	PAM-4	**DMT**
Analog EQ	CTLE	CTLE	CTLE	CTLE	**–**
ADC Res. (bits)	8	7	6	7	**10/8***
Fs (GS/s)	28	28	32.1875	28	**22.4**
DSP Power (mW)	N.A.	220	N.A.	N.A.	**68.22**
RX Tot. Power (mW)	280**	545	283.9**	180****	**161.46**
Energy (pJ/b)	5**	9.7	4.41**	3.21****	**2.88**
Supplies (V)	0.9/1.2/1.8	0.85/0.9/1.2/1.8	0.9/1.2	0.8/1	**0.8/0.7/0.7/0.75**
RX Area (mm2)	N.A.**	1.76***	0.1625**	0.32****	**0.26**
Ch. Loss (dB)	31	32	29.5	40	**CH-A:11***** **CH-B:18***** **CH-C:28*****
EQ. tap adaptation	On-chip	On-chip	Off-chip	On-chip	**Off-chip**
Pre-FEC BER	<1e-8	<1e-12	<1e-4	<1e-5	**CH-A:<1e-6 CH-B:<1e-6 CH-C:<2e-4**
Technology	16nm FinFET	16nm FinFET	16nm FinFET	7nm FinFET	**14nm FinFET**

*DSP receives 8 MSBs for equalization out of 10 bits.
**DSP is not implemented in silicon; hence DSP area and power are not included.
***Estimated from die photo reported in [6].
****Includes whole TRX analog, excludes DSP.
*****Loss at fNYQ of PAM-4 equivalent system for comparison purpose. Actual losses in DMT system are 8dB(CH_A), 13dB(CH_B) and 20dB(CH_C).

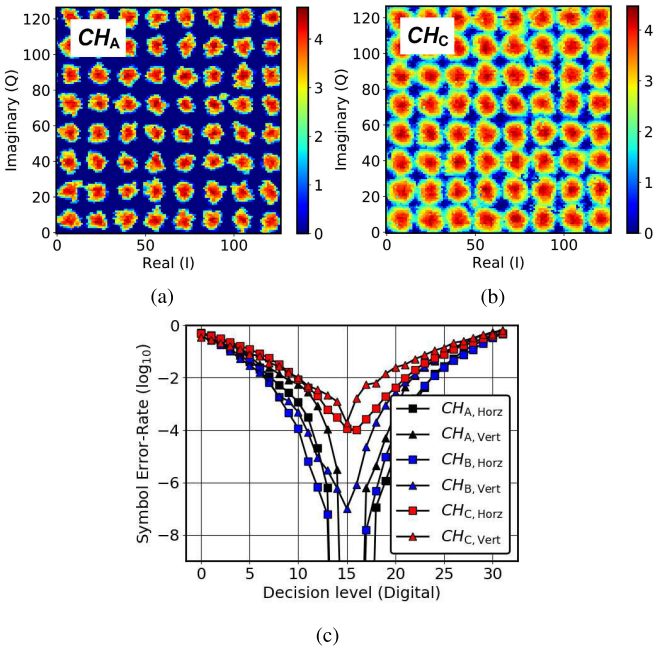


Fig. 16. Equalized constellation diagrams (top row) for (a) CH_A and (b) CH_C with sub-channels' power adjustment, and (c) bathtub curves for $CH_{A/B/C}$.

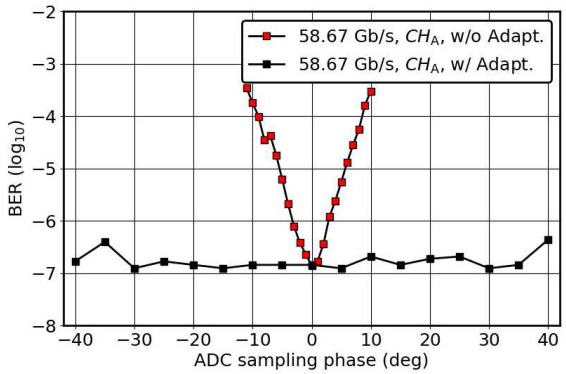


Fig. 17. BER versus ADC sampling phase with and without FDE's coefficient adaptation.

consumed in the DSP), corresponding to an energy efficiency of 2.9 pJ/b (1.2 pJ/b for the DSP).

The BER with respect to the ADC sampling skew was measured using the on-chip BER counter. Since a residual delay in time-domain translates into rotation in phase, the overall constellation diagram is rotated as a sampling delay is introduced. Additionally the amplitude of the digitized samples scales with the ADC sampling phase shift from the optimal phase, resulting in an inaccurate amplitude correction of the constellation diagrams unless the equalization coefficients are properly adjusted. Fig. 17 shows the bit-error count results with 58.67 Gb/s (64-QAM for 1st-14th tone, 16-QAM for the 15th tone) for CH_A, versus the ADC sampling skew with and without coefficient adaptation. The measurement shows how a properly adjusted digital equalizer can fully compensate for any non-integer time offset, which relaxes synchronization requirements. It can also be noted from Fig. 17 that a random sampling clock jitter can cause a random rotation of the constellation diagram that cannot be corrected by equalizer adjustment, affecting the BER performance. The DAC/ADC

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sampling rate is increased from 22.4 to 24 GS/s for this measurement. To this end, nominal supply voltages are all increased to 0.8 V for a reliable operation, resulting in a decreased RX energy efficiency of 3.66 pJ/b for the 58.67-Gb/s data-rate target. The pre-forward error correction (FEC) BER is measured to be lower than 1e-6 for CH_{A} over a range of 40° of the sampling phase skew from the optimal sampling phase.

Table I summarizes the performance of the implemented DMT RX prototype and compares it with other state-of-the-art ADC-based RXs having similar data-rates. The pre-FEC BER is not as low as that of [4] and [6] with similar channel loss, while the presence of the CTLE would potentially improve the BER performance. This work exhibits the highest energy efficiency among compared prior arts and requires only 22.4 GS/s of DAC/ADC conversion rate, which is 20% lower than 56-Gb/s PAM-4 systems.

V. Conclusion

This article presents a 56-Gb/s electrical serial-data RX employing DMT modulation implemented in 14-nm FinFET process. The work shows how more complex digital modulation techniques together with a digital receiver can achieve high data rates by using high spectral efficiency modulation. Without the presence of a CTLE, the receiver is able to achieve <2e-4 and <1e-6 raw BER over a channel with 28-dB loss and channels with <18 dB loss at 14 GHz, respectively. The measured receiver energy efficiency is better than 3 pJ/b including the ADC and the DSP together with test circuits. The DSP dissipates only 68 mW, which is significantly lower than existing ADC-based PAM-4 receiver DSPs.

Acknowledgment

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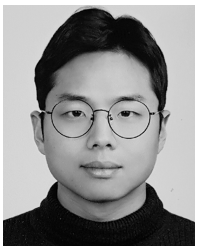
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